

ANSWER SHEET (MIDSEM) DL

① (a) False, zero training error means overfitting; however bias can be there due to model assumptions with respect to the true function.

(b) False, Increasing the model capacity increases the risk of overfitting. However, in deep learning models in particular, regularization, large data, and training strategies can still allow good generalization.

(c) True, Adam combines Momentum and RMSprop ideas.

(d) False, Although less severe as compared to sigmoid/tanh, vanishing gradient can still occur in neural networks with ReLU activations, when neurons become inactive (dead ReLU).

(e) True, Momentum helps optimization by reducing oscillations caused by anisotropic curvature in the loss surface and accelerating progress along shallow directions.

② (a) Since the true data distribution is unknown, machine learning minimizes the average loss over the training dataset (empirical risk)

$$R_{\text{emp}}(f) = \frac{1}{N} \sum_{i=1}^N \ell(y_i, f(x_i))$$

$N \rightarrow$ No. of training samples

(b)

or

$$\hat{R}(f, d) = \frac{1}{N} \sum_{(x_i, y_i) \in d} \ell(y_i, f(x_i))$$

d is the training dataset and N is the no. of training samples.

(b) $E[(y - f(x))^2] = \text{Bias}^2 + \text{Variance} + \text{noise}$

* The noise term quantifies the irreducible part of the expected risk.

* The Bias term measures the discrepancy b/w the avg. model and the Bayes model.

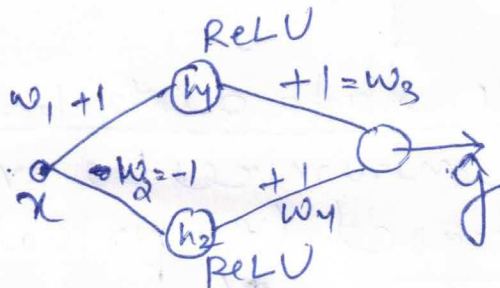
* The Variance term quantifies the variability of the prediction.

(c) A feed-forward nn with a single hidden layer ^(containing finite no. of neurons) and a non-linear activation function can approximate any continuous function on a compact domain.

(d) $f(x) = \text{sign}\left(\sum_i w_i x_i + b\right)$

where $\text{sign}(z) = \begin{cases} 1, & \text{if } z \geq 0 \\ 0, & \text{otherwise} \end{cases}$

(e) we can write $|x| = \text{ReLU}(x) + \text{ReLU}(-x)$



$$h_1 = \text{ReLU}(x), \quad w_1 = +1, \quad b_1 = 0$$

$$h_2 = \text{ReLU}(-x), \quad w_2 = -1, \quad b_2 = 0$$

Prediction $\hat{y} = h_1 + h_2$, $w_3 = +1, w_4 = +1, b = 0$.

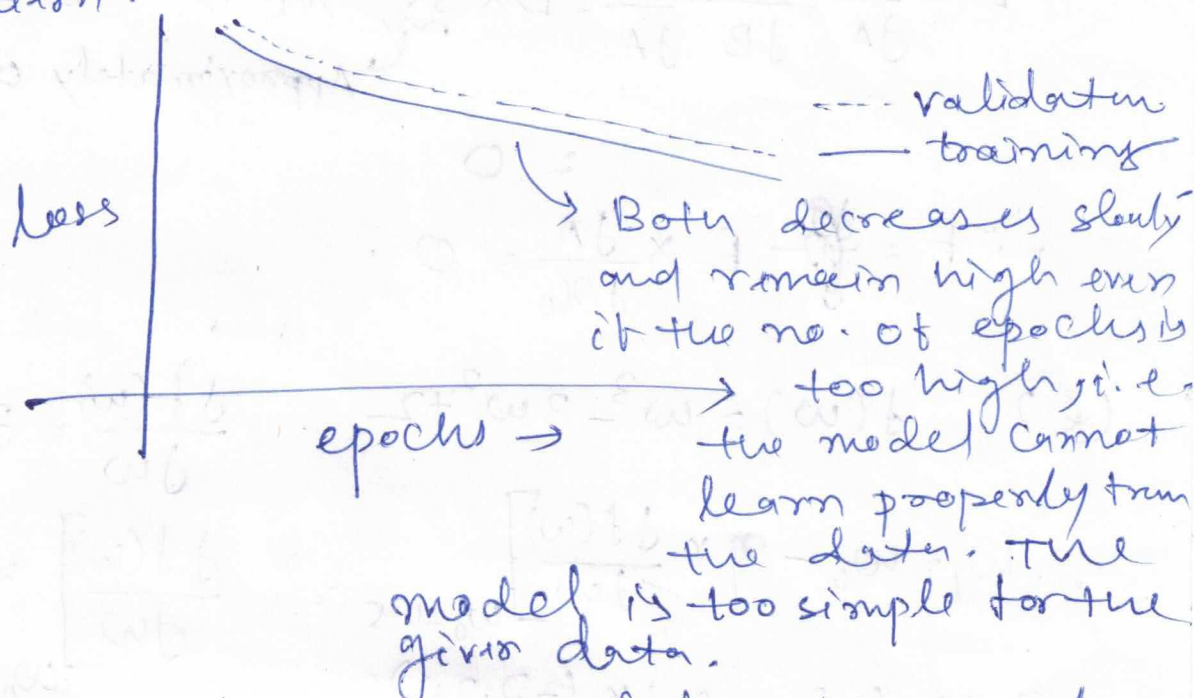
③ (a) Using momentum optimizer -
 velocity $u_t = \alpha u_{t-1} - \eta g_t$
 $Q_t = Q_{t-1} + u_t$

Here g_t is the gradient considering the parameters in $(t-1)$ th iteration.

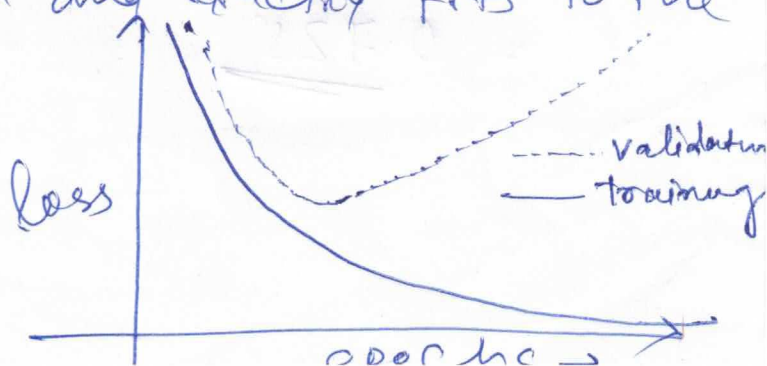
$$\therefore u_t = 0.9 \begin{bmatrix} 0.2 \\ -0.1 \end{bmatrix} - 0.1 \times \begin{bmatrix} 0.6 \\ -0.4 \end{bmatrix} = \begin{bmatrix} 0.12 \\ -0.05 \end{bmatrix}$$

$$\therefore Q_t = \begin{bmatrix} 1.5 \\ -2.0 \end{bmatrix} + \begin{bmatrix} 0.12 \\ -0.05 \end{bmatrix} = \begin{bmatrix} 1.62 \\ -2.05 \end{bmatrix}$$

(b) In case of underfitting, both training and validation losses remain high and decrease very slowly because the model cannot learn from the data pattern.



In case of overfitting, the model is too complex for the given data. The model starts to learn the noise in the training data and exactly fits to the training data giving low training loss. However, the validation loss decreases initially, then starts to increase again.

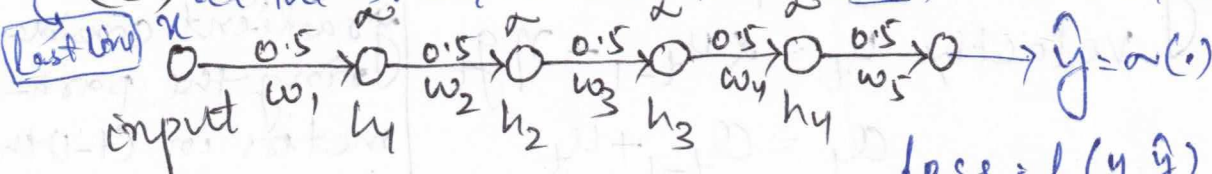


(c) yes, it shows VG problem as the gradient at the 1st layer is $(0.1)^5 = 0.00001$, which is very small

Diagram illustrating the vanishing gradient problem in a deep neural network with 5 layers (input, 4 hidden, and output).

The diagram shows the flow of information from input to output, with weights w_1, w_2, w_3, w_4, w_5 and hidden layer nodes h_1, h_2, h_3, h_4 . The output is $\hat{y} = \sigma(\cdot)$.

The loss is calculated as $\text{loss} = l(y, \hat{y})$.



$$L_{SS} = L(y, \hat{y})$$

Gradient at the first layer is

given by $\frac{dL}{dw_1} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dw_1} = \frac{dL}{d\hat{y}} \cdot \frac{dw_4}{dw_1} \cdot \frac{dw_3}{dw_4} \cdot \frac{dw_2}{dw_3} \cdot \frac{dw_1}{dw_2}$

$\frac{\partial L}{\partial g} = 1$ as the gradient at the o/p layer is 1

$$= 1 \times (u)^5 \times \sigma^{-1}(\cdot) \times \sigma^{-1}(\cdot) \times \sigma^{-1}(\cdot) \times \sigma^{-1}(\cdot) \times \sigma^{-1}(\cdot) \quad \boxed{\text{Step 1}}$$

$$= 1 \times (u)^5 \times (\sigma^{-1}(\cdot))^5 = (1 \times 1) \times (0.2)^5 = (0.2)^5 \quad \boxed{\text{Answer}}$$

↳ as the derivative of the
at each layer is 0.2

(d) $D = \frac{\partial L}{\partial B} = \frac{\partial L}{\partial C} \frac{\partial C}{\partial B} = 0.63 \times C(1-C) \quad n_0 = 9$

$$= 0.63 \times 0.88 \times 0.12 \quad \checkmark A = 3 \times 10^2 = 243$$

0.066

$$\checkmark B = 3e^{-A} + 2 \approx 2$$

$$\checkmark E = \frac{\partial L}{\partial A} = \frac{\partial L}{\partial B} \cdot \frac{\partial B}{\partial A} = D \times \underbrace{3e^{-A}}_{\text{Approximately 0}} \times (-1) \quad \checkmark C = \sigma(B) = \sigma(2) = 0.88$$

100

$$\therefore F = \cancel{E} \times \frac{\partial A}{\partial x_0} = 0$$

(e) $f(\omega) = \omega^3 - 3\omega^2 + 2$

$$\frac{df(w)}{dw} = 3w^2 - 6w$$

$$\therefore \omega_1 = \omega_0 - \eta \times \left[\frac{df(\omega)}{d\omega} \right]_{\omega_0=0.5} \quad \therefore \left[\frac{df(\omega)}{d\omega} \right]_{\omega_0=0.5} = -2.25$$

$$= 0.5 - 0.1 \times (-2.25)$$

$$= 0.5 - 0.1 \times (-2.25)$$

$$= 0.5 + 0.0225 = \cancel{0.5225}$$

$$= \underline{\underline{0.725}}$$

④ (a) Single Layer Perceptron

In regression, $\hat{y} = W^T X$ for a single layer perceptron.

Here, $W = [w_1, w_2, w_3, b]^T$ and $X = [x_1, x_2, x_3, 1]^T$

→ Using mini-batch SGD,

$$Q_{t+1} = Q_t - \eta (\nabla_Q(Q_t))$$

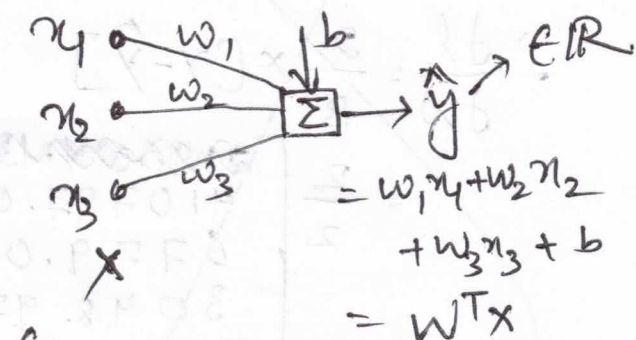
$$= Q_t - \eta \left(\frac{\partial L}{\partial Q} \right)_{\text{at } Q_t}$$

Here $L = \frac{1}{B} \sum_{i=1}^B (y_i - \hat{y}_i)^2$

* Squared error is used as it is a regression problem.

Here $Q = \{W\} = \{w_1, w_2, w_3, b\}$

$$\therefore \frac{\partial L}{\partial Q} = \frac{\partial L}{\partial W} = -\frac{2}{B} \sum_{i=1}^B (y_i - \hat{y}_i) X_i = -\frac{2}{B} X [Y - \hat{Y}]$$



(No activation function is required, as it is a regression problem)

Prediction vector corresponding to a batch

{Batch-1: {Sample 1, Sample 2}}
{Batch-2: {Sample 3, Sample 4}}
X is the matrix made out of batches
Y is the target vector corresponding to a batch

→ We can take random samples in batches.

Batch-1:

$$X = \begin{bmatrix} 102.5 & 98.2 \\ 1.8 & 2.1 \\ 0.75 & 0.60 \\ 1 & 1 \end{bmatrix} \quad \begin{matrix} \text{sample-1} \\ \text{sample-2} \end{matrix}$$

$$Y - \hat{Y} = \begin{bmatrix} 105.4 - 10.81 \\ 100.8 - 10.33 \end{bmatrix} = \begin{bmatrix} 94.59 \\ 90.47 \end{bmatrix}$$

$$\therefore \frac{\partial L}{\partial W} = -\frac{2}{B} X [Y - \hat{Y}] = -\frac{2}{2} \begin{bmatrix} 18579.6 \\ 360.2 \\ 125.2 \\ 185.1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ b \end{bmatrix} = \begin{bmatrix} 0.10 \\ -0.05 \\ 0.2 \\ 0.5 \end{bmatrix} - 0.001 \times \frac{\partial L}{\partial W} = \begin{bmatrix} 0.10 \\ -0.05 \\ 0.20 \\ 0.5 \end{bmatrix} + \begin{bmatrix} 18.58 \\ 0.36 \\ 0.125 \\ 0.1851 \end{bmatrix}$$

updated weights after first batch

$$\therefore \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ b \end{bmatrix} = \begin{bmatrix} 18.68 \\ 0.31 \\ 0.33 \\ 0.69 \end{bmatrix}$$

Batch-2!

$$X = \begin{bmatrix} 105.7 & 99.4 \\ 1.5 & 1.9 \\ 0.82 & 0.7 \\ 1 & 1 \end{bmatrix} \quad \begin{matrix} \nearrow \text{Sample-3} \\ \nearrow \text{Sample-4} \end{matrix}$$

4×2

$$y - \hat{y} = \begin{bmatrix} 108.9 - 1975.9 \\ 102.3 - 1858.3 \end{bmatrix}$$

$$\therefore \frac{\partial L}{\partial w} = -\frac{2}{B} \times [y - \hat{y}]$$

$$= -\frac{2}{2} \begin{bmatrix} -371888.3 \\ -6136.9 \\ -2760.14 \\ -3623 \end{bmatrix} = \begin{bmatrix} 371888.3 \\ 6136.9 \\ 2760.14 \\ 3623 \end{bmatrix}$$

2×1

$$\therefore \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ b \end{bmatrix} = \begin{bmatrix} 18.68 \\ 0.31 \\ 0.33 \\ 0.69 \end{bmatrix} - 0.001 \times \frac{\partial L}{\partial w} = \begin{bmatrix} 18.68 \\ 0.31 \\ 0.33 \\ 0.69 \end{bmatrix} - \begin{bmatrix} 371.89 \\ 6.14 \\ 2.76 \\ 3.62 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ b \end{bmatrix} = \begin{bmatrix} -353.21 \\ -5.83 \\ -2.43 \\ -2.93 \end{bmatrix}$$

(updated weights after one epoch)

* will get full marks if you have completed till this point

$$\text{Loss} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

$$\approx 1.29 \times 10^9$$

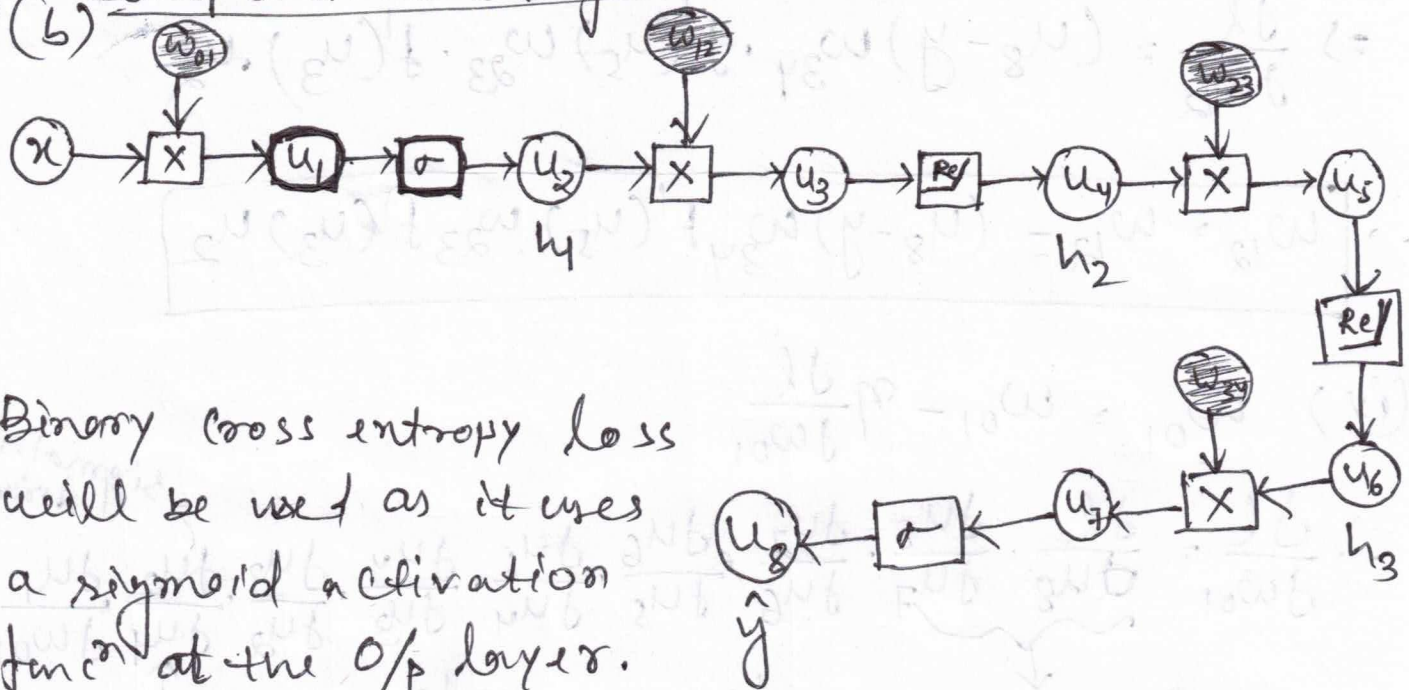
$$\hat{y}_1 = -36204.92$$

$$\hat{y}_2 = -34698.54$$

$$\hat{y}_3 = -37341.57$$

$$\hat{y}_4 = -35093.18$$

Computational Diagram



Binary cross entropy loss will be used as it uses a sigmoid activation function at the o/p layer.

$$l = -[y \log(\hat{y}) + (1-y) \log(1-\hat{y})]$$

$$(i) w_{34} = w_{34} - \eta \frac{\partial l}{\partial w_{34}}$$

$$\frac{\partial l}{\partial w_{34}} = \frac{\partial l}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial u_7} \cdot \frac{\partial u_7}{\partial w_{34}}$$

$$= (u_8 - y) u_6$$

$$\therefore w_{34} = w_{34} - \eta (u_8 - y) u_6$$

$$(ii) w_{23} = w_{23} - \eta \frac{\partial l}{\partial w_{23}}$$

$$\frac{\partial l}{\partial w_{23}} = \frac{\partial l}{\partial u_8} \cdot \frac{\partial u_8}{\partial u_7} \cdot \frac{\partial u_7}{\partial u_6} \cdot \frac{\partial u_6}{\partial u_5} \cdot \frac{\partial u_5}{\partial w_{23}}$$

$$= (u_8 - y) w_{34} \cdot f'(u_5) \cdot u_4$$

$$\therefore w_{23} = w_{23} - \eta (u_8 - y) w_{34} f'(u_5) \cdot u_4$$

$$(iii) w_{12} = w_{12} - \eta \frac{\partial l}{\partial w_{12}}$$

$$\frac{\partial l}{\partial w_{12}} = \frac{\partial l}{\partial u_8} \cdot \frac{\partial u_8}{\partial u_7} \cdot \frac{\partial u_7}{\partial u_6} \cdot \frac{\partial u_6}{\partial u_5} \cdot \frac{\partial u_5}{\partial u_4} \cdot \frac{\partial u_4}{\partial u_3} \cdot \frac{\partial u_3}{\partial w_{12}}$$

$$u_1 = w_{01} x$$

$$h_1 = u_2 = \sigma(u_1)$$

$$u_3 = w_{12} u_2$$

$$h_2 = u_4 = \text{Re}(u_3) = f(u_3)$$

$$u_5 = w_{23} u_4$$

$$h_3 = u_6 = \text{Re}(u_5) = f(u_5)$$

$$u_7 = w_{34} u_6$$

$$\hat{y} = u_8 = \sigma(u_7)$$

In general,

$$f'(a) = 0, \text{ if } a \leq 0$$

$$= 1, \text{ if } a > 0$$

$$(f'(u_5) = 0, \text{ if } u_5 \leq 0)$$

$$= 1, \text{ if } u_5 > 0)$$

$$\Rightarrow \frac{\partial l}{\partial w_{12}} = (u_8 - y) w_{34} \cdot f'(u_5) w_{23} \cdot f'(u_3) \cdot u_2$$

$$\therefore \boxed{w_{12} = w_{12} - (u_8 - y) w_{34} f'(u_5) w_{23} f'(u_3) u_2}$$

$$(iv) \quad w_{01} = w_{01} - \eta \frac{\partial l}{\partial w_{01}}$$

$$\begin{aligned} \therefore \frac{\partial l}{\partial w_{01}} &= \frac{\partial l}{\partial u_8} \cdot \frac{\partial u_8}{\partial u_7} \cdot \frac{\partial u_7}{\partial u_6} \cdot \frac{\partial u_6}{\partial u_5} \cdot \frac{\partial u_5}{\partial u_4} \cdot \frac{\partial u_4}{\partial u_3} \cdot \frac{\partial u_3}{\partial u_2} \cdot \frac{\partial u_2}{\partial u_1} \cdot \frac{\partial u_1}{\partial w_{01}} \\ &= (u_8 - y) \cdot w_{34} \cdot f'(u_5) \cdot w_{23} \cdot f'(u_3) \cdot w_{12} \cdot \underbrace{u_2(1-u_2)}_{\text{sigmoidal derivative}} \cdot x \end{aligned}$$

$$\therefore \boxed{w_{01} = w_{01} - (u_8 - y) w_{34} f'(u_5) w_{23} f'(u_3) w_{12} u_2 (1-u_2) x}$$

* Most of the terms are repeated from previous iterations. Just the last two terms need to be derived.

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